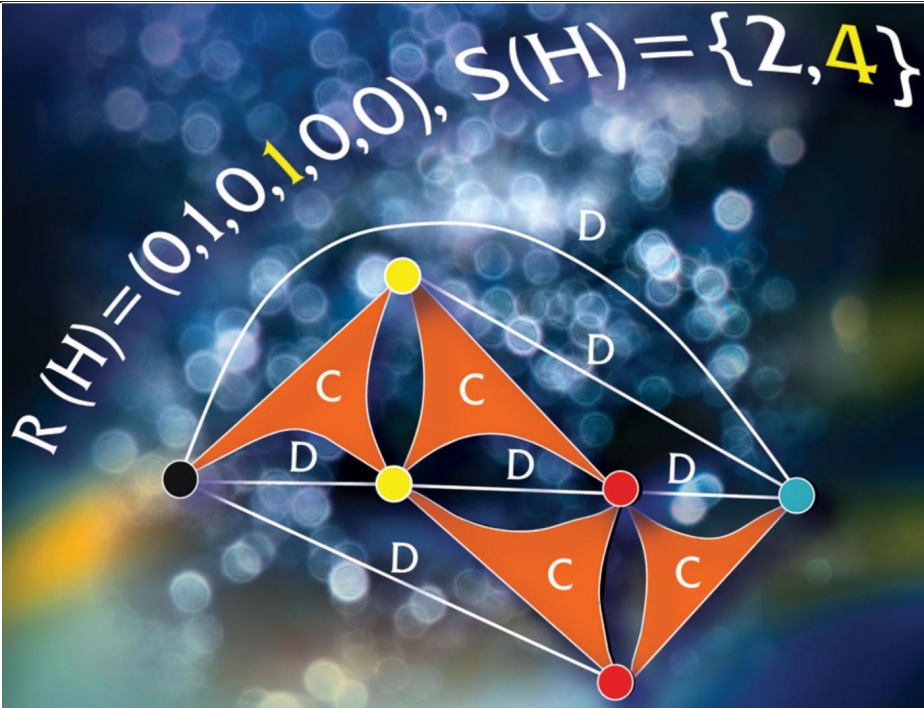
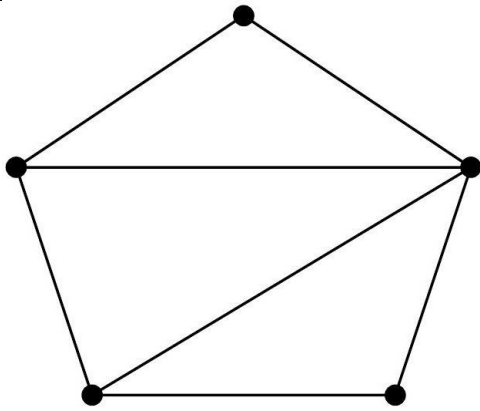
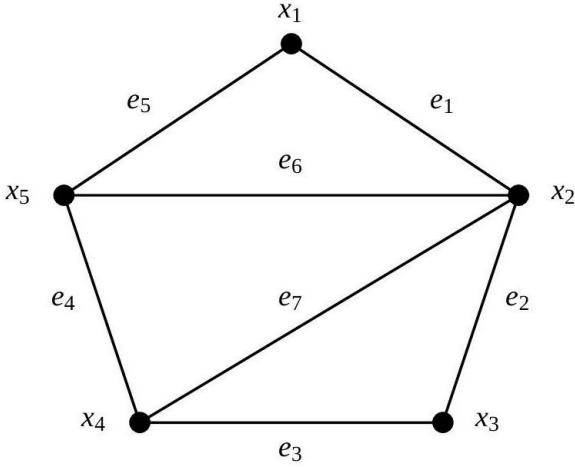
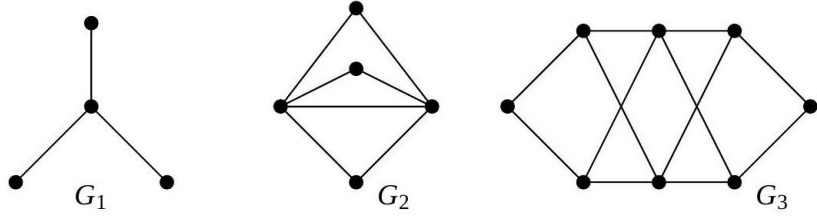
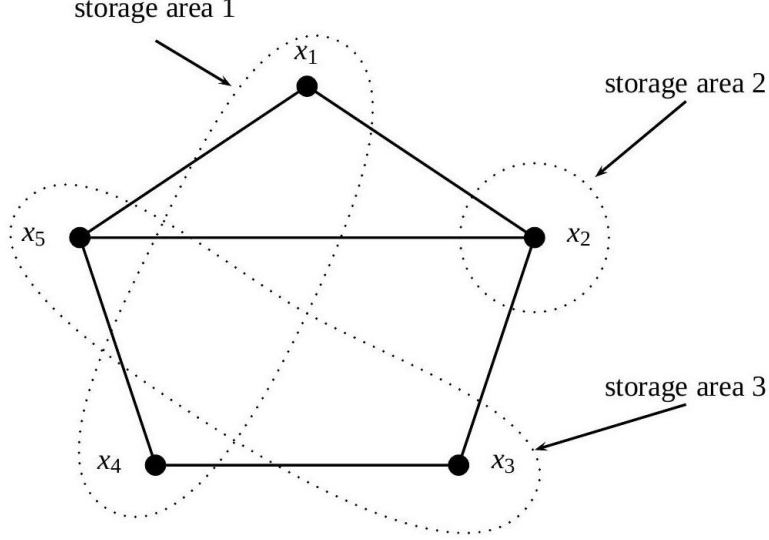
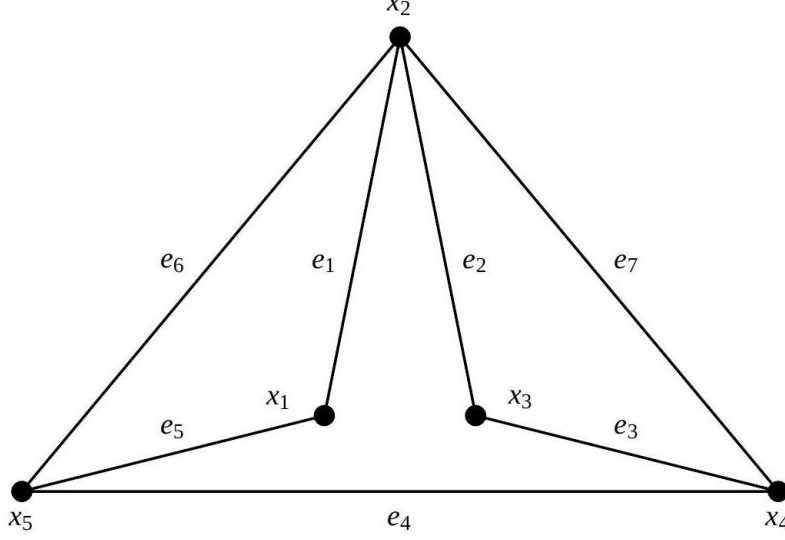
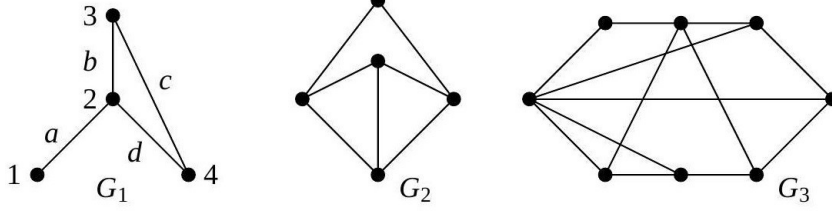
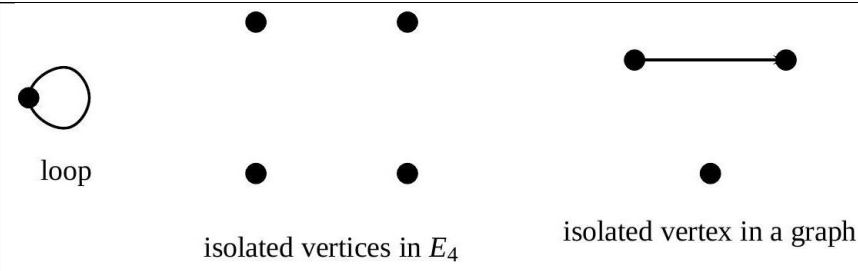
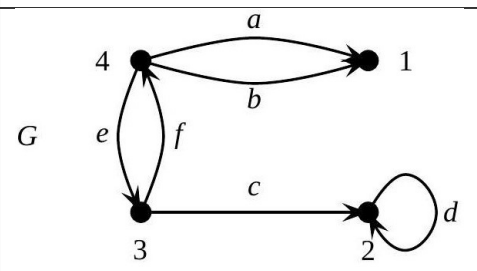
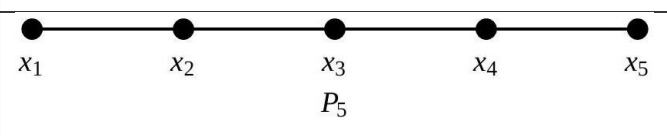
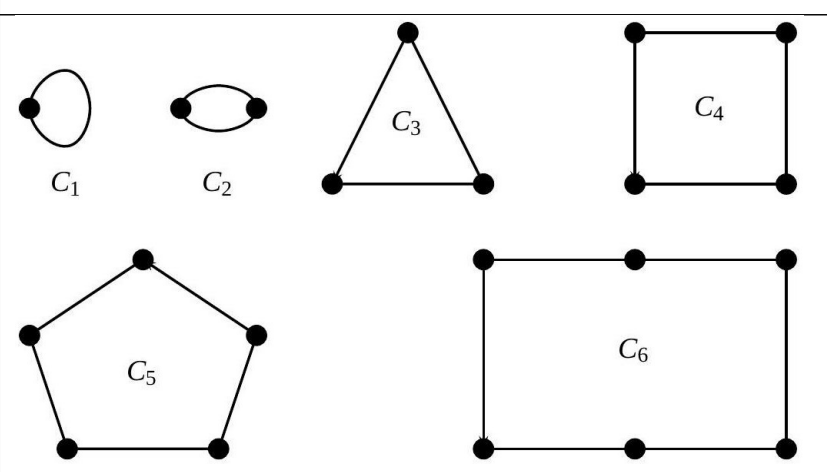
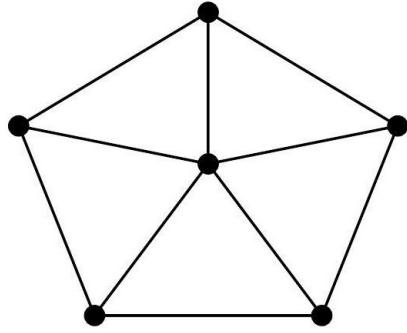
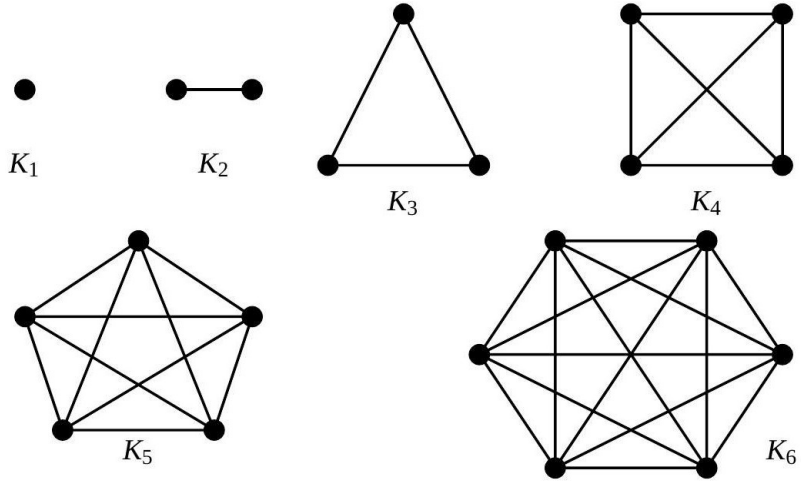
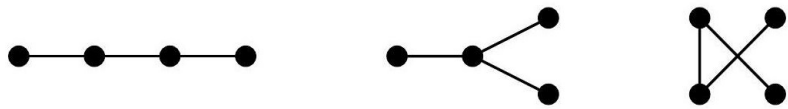
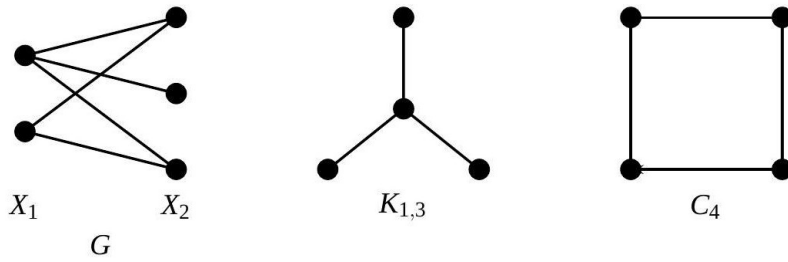
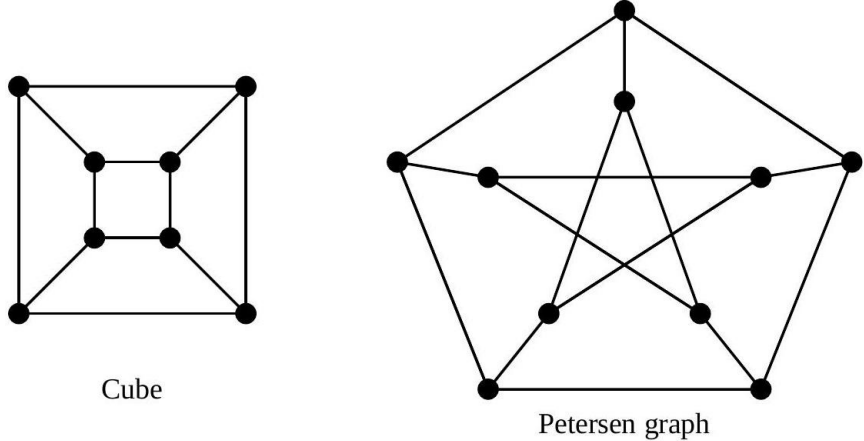


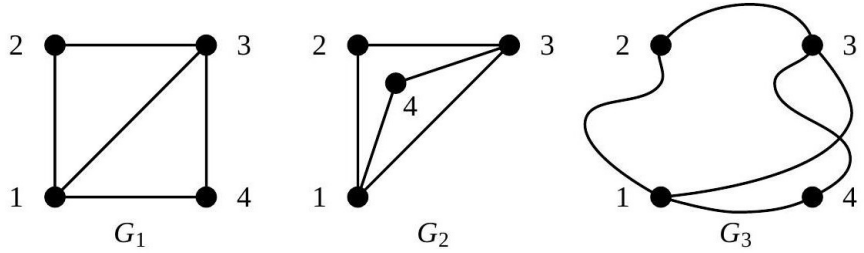
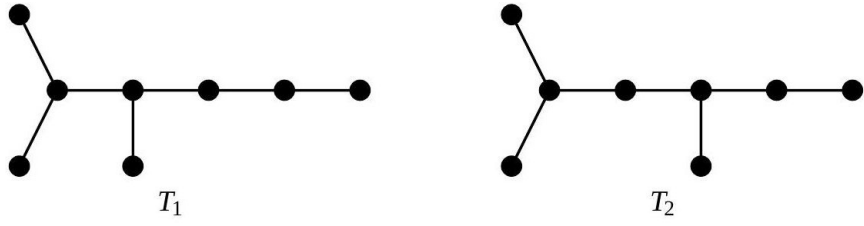
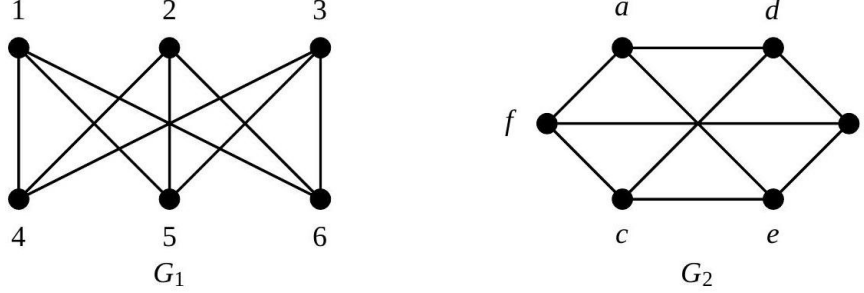
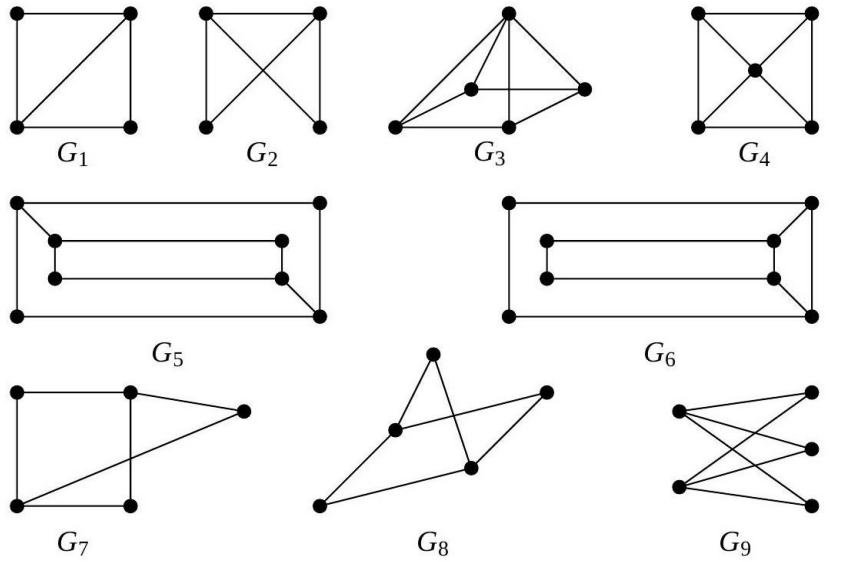
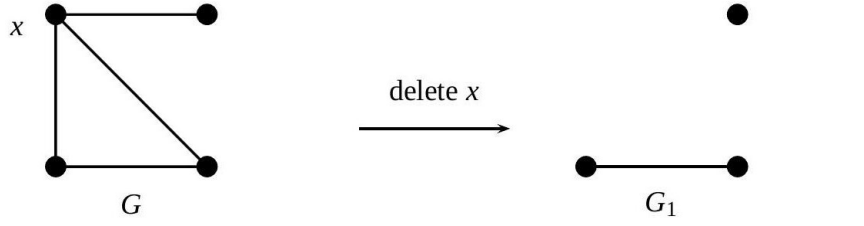
Image regions

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voloshin-hypergraph_ DIA_0006	29	—	—	—	
voloshin-hypergraph_ DIA_0007	31	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0008	32	—	—	—	 loop isolated vertices in E_4 isolated vertex in a graph multiple edges arcs
voloshin-hypergraph_ DIA_0009	32	—	—	—	
voloshin-hypergraph_ DIA_0010	33	—	—	—	 G
voloshin-hypergraph_ DIA_0011	34	—	—	—	 P_5
voloshin-hypergraph_ DIA_0012	34	—	—	—	 G_1 G_2
voloshin-hypergraph_ DIA_0013	35	—	—	—	 C_1 C_2 C_3 C_4 C_5 C_6

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voloshin-hypergraph_ DIA_0017	37	—	—	—	
voloshin-hypergraph_ DIA_0018	38	—	—	—	

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voloshin-hypergraph_ DIA_0021	40	—	—	—	
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voloshin-hypergraph_ DIA_0023	42	—	—	—	

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voloshin-hypergraph_ DIA_0026	43	—	—	—	
voloshin-hypergraph_ DIA_0027	44	—	—	—	
voloshin-hypergraph_ DIA_0028	44	—	—	—	

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voloshin-hypergraph_ DIA_0031	48	—	—	—	
voloshin-hypergraph_ DIA_0032	49	—	—	—	
voloshin-hypergraph_ DIA_0033	51	—	—	—	

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voloshin-hypergraph_ DIA_0037	53	—	—	—	
voloshin-hypergraph_ DIA_0038	56	—	—	—	

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voloshin-hypergraph_ DIA_0040	59	—	—	—	
voloshin-hypergraph_ DIA_0041	60	—	—	—	
voloshin-hypergraph_ DIA_0042	61	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0043	64	—	—	—	Diagram illustrating a hypergraph structure. Two sets, X and Y, are shown. X contains S and $X - S_0$. Y contains $N_G(S)$ and $N_G(S_0)$. The region between S_0 and $N_G(S_0)$ is labeled G_A. The region between S and $N_G(S)$ is labeled G_B. Lines connect S to $N_G(S_0)$ and S_0 to $N_G(S)$.
voloshin-hypergraph_ DIA_0044	65	—	—	—	Diagram illustrating a hypergraph structure. Two sets, X and Y, are shown. X contains S and A. Y contains $N_{G_1}(S)$ and B. The region between S and $N_{G_1}(S)$ is labeled G_1. The region between A and B is labeled G_2. A dashed line separates G_1 and G_2.
voloshin-hypergraph_ DIA_0045	66	—	—	—	Diagram illustrating a grid structure. A set of points is arranged in a grid, connected by lines, forming a rectangular structure.

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voloshin-hypergraph_ DIA_0046	68	—	—	—	
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voloshin-hypergraph_ DIA_0050	74	—	—	—	
voloshin-hypergraph_ DIA_0051	75	—	—	—	

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voloshin-hypergraph_ DIA_0056	78	—	—	—	

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voloshin-hypergraph_ DIA_0059	81	—	—	—	
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voloshin-hypergraph_ DIA_0062	84	—	—	—	
voloshin-hypergraph_ DIA_0063	86	—	—	—	
voloshin-hypergraph_ DIA_0064	87	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0065	88	—	—	—	The diagram illustrates the construction of a hypergraph from two base graphs, K_5 and $K_{3,3}$. Top section (K_5): A complete graph K_5 with vertices 1, 2, 3, 4, 5 is shown. It is transformed into a triangle with vertices 1, 2, 3. Vertex 4 is placed either 'inside' or 'outside' the triangle. In both cases, the resulting graph has 'no face for 5'. Bottom section ($K_{3,3}$): A complete bipartite graph $K_{3,3}$ with vertices 1, 2, 3, 4, 5, 6 is shown. It is transformed into a square with vertices 1, 2, 3, 4. Vertex 5 is placed either 'inside' or 'outside' the square. In both cases, the resulting graph has 'no face for 6'.
voloshin-hypergraph_ DIA_0066	91	—	—	—	Two graphs, G_1 and G_2, are shown. G_1 is a complex graph with 10 vertices and many edges. G_2 is a graph with 10 vertices and 12 edges, including a cycle of length 6.
voloshin-hypergraph_ DIA_0067	92	—	—	—	A graph with 10 vertices and 12 edges, including a cycle of length 6.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0068	93	—	—	—	
voloshin-hypergraph_ DIA_0069	97	—	—	—	
voloshin-hypergraph_ DIA_0070	99	—	—	—	
voloshin-hypergraph_ DIA_0071	100	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0072	101	—	—	—	Diagram illustrating the decomposition of a hypergraph G into two hypergraphs G_1 and G_2. G consists of two vertices x and y connected to a set of three internal vertices. G_1 is identical to G but with a dashed edge between x and y. G_2 consists of a single vertex xy connected to the same three internal vertices. The equation $G = G_1 + G_2$ is shown.
voloshin-hypergraph_ DIA_0073	102	—	—	—	Diagram illustrating the decomposition of a tree G into two components G_1 and G_2. G is a tree with vertices x, y, and z. $G_1 = K_3$ is a triangle with vertices x, y, and z. $G_2 = K_2$ is a single edge between vertices x and z. The equation $G = G_1 + G_2$ is shown.
voloshin-hypergraph_ DIA_0074	103	—	—	—	Diagram illustrating a hierarchical tree structure G. The root node G branches into G_1 and G_2. G_1 branches into G_3 and G_4, and G_2 branches into G_5 and G_6. Each of these nodes further branches into two leaf nodes. Below the tree, a sequence of components $K_{i_1}, K_{i_2}, \dots, K_{i_s}$ is shown, representing the final decomposition of the tree.

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voloshin-hypergraph_ DIA_0075	104	—	—	—	
voloshin-hypergraph_ DIA_0076	105	—	—	—	
voloshin-hypergraph_ DIA_0077	107	—	—	—	

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voloshin-hypergraph_ DIA_0078	108	—	—	—	
voloshin-hypergraph_ DIA_0079	109	—	—	—	
voloshin-hypergraph_ DIA_0080	111	—	—	—	
voloshin-hypergraph_ DIA_0081	112	—	—	—	
voloshin-hypergraph_ DIA_0082	113	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0083	115	—	—	—	A diagram illustrating a hypergraph structure. It shows three sets X_1, X_2, and X_k represented as ovals. A point x_0 is connected to X_1 and X_2 by solid lines. A dashed line connects x_0 to X_k. Points p_1 and p_2 are marked on the lines connecting x_0 to X_1 and X_2 respectively. A bracket labeled A is shown below the points.
voloshin-hypergraph_ DIA_0084	117	—	—	—	A diagram of a graph G with 8 vertices labeled 1 through 8. The vertices are arranged in two rows: 1, 4, 5, 7 in the front and 6, 3, 2, 8 in the back. Edges connect (1,4), (1,5), (4,6), (4,5), (5,7), (6,7), (6,3), (7,8), (3,8), (3,2), and (8,2). There are also dashed curved lines connecting 1 to 5 and 4 to 6.
voloshin-hypergraph_ DIA_0085	119	—	—	—	A diagram showing a large oval labeled G_{i+1} and a smaller oval labeled $N(x_i)$ inside it. A point x_i is outside G_{i+1}, and two lines connect it to the boundary of $N(x_i)$. A bracket labeled G_i is shown below the large oval.
voloshin-hypergraph_ DIA_0086	120	—	—	—	A diagram of a graph with vertices labeled a, b, c, d, e, x and several unlabeled vertices. Vertex x is a central node connected to a, b, c, d, e. Other vertices are connected in a complex structure: a to 1, 3, 4; b to 2, 4; c to 3, 1; d to 4; e to 5. There are also edges between 1 and 3, 2 and 4, 3 and 1, 4 and 2, 3 and 1, 4 and 2, 3 and 1, 4 and 2.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0087	121	—	—	—	
voloshin-hypergraph_ DIA_0088	122	—	—	—	
voloshin-hypergraph_ DIA_0089	124	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0090	125	—	—	—	The diagram shows a transformation. On the left, a single vertical edge connects two vertices. An arrow points to the right, where a more complex graph is shown. This graph has five vertices. It consists of a central diamond shape formed by four vertices, with a fifth vertex at the top. The top vertex is connected to all four vertices of the diamond. The two vertices on the left of the diamond are connected to each other, and the two vertices on the right are connected to each other. A dotted ellipse encircles the two vertices on the right.
voloshin-hypergraph_ DIA_0091	125	—	—	—	The diagram shows a complex graph with 10 vertices. It has a central vertex connected to four other vertices, which are arranged in a square. Each of these four vertices is further connected to two more vertices, forming a total of 10 vertices. The edges form a dense, symmetric pattern.
voloshin-hypergraph_ DIA_0092	126	—	—	—	The diagram shows two graphs, G and $\bar{G}$. Graph G on the left has five vertices labeled 1, 2, 3, 2, 3. It is a planar graph with edges connecting the vertices in a specific way. Graph $\bar{G}$ on the right is the complement of G, with the same five vertices and edges connecting the non-adjacent vertices of G. Both graphs are enclosed in dotted ellipses.
voloshin-hypergraph_ DIA_0093	129	—	—	—	The diagram shows two rows of graphs. The top row shows graph G on the left and its line graph $L(G)$ on the right. Graph G has four vertices labeled a, b, c, d and edges $a-b, b-c, c-d, d-a, b-d, c-b$. Graph $L(G)$ has four vertices labeled a, b, c, d and edges $a-b, b-c, c-d, d-a, b-d, c-b$. The bottom row shows graph G on the left and its line graph $L(G)$ on the right. Graph G has four vertices labeled 1, 2, 3, 4 and edges 1-2, 2-3, 3-4, 4-1, 2-4, 3-1. Graph $L(G)$ has four vertices labeled 1, 2, 3, 4 and edges 1-2, 2-3, 3-4, 4-1, 2-4, 3-1. The text 'edge coloring' is below the first row and 'vertex coloring' is below the second row.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0094	130	—	—	—	
voloshin-hypergraph_ DIA_0095	131	—	—	—	
voloshin-hypergraph_ DIA_0096	133	—	—	—	
voloshin-hypergraph_ DIA_0097	134	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0098	136	—	—	—	G
voloshin-hypergraph_ DIA_0099	140	—	—	—	
voloshin-hypergraph_ DIA_0100	141	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0101	145	—	—	—	The image displays three network flow diagrams arranged vertically, connected by downward arrows, illustrating the progression of a flow augmentation algorithm. Each diagram features a source node u and a sink node v, with intermediate nodes x_1, x_2, x_3, x_4. Initial flow F_0: - Edges and flows: $u \rightarrow x_1$ (0,3), $u \rightarrow x_4$ (1,5), $x_1 \rightarrow x_2$ (1,7), $x_1 \rightarrow x_3$ (1,1), $x_2 \rightarrow v$ (1,6), $x_3 \rightarrow v$ (0,2), $x_4 \rightarrow x_3$ (1,4). Flow F_1: - Edges and flows: $u \rightarrow x_1$ (1,3), $u \rightarrow x_4$ (1,5), $x_1 \rightarrow x_2$ (1,7), $x_1 \rightarrow x_3$ (0,1), $x_2 \rightarrow v$ (1,6), $x_3 \rightarrow v$ (1,2), $x_4 \rightarrow x_3$ (1,4). Flow F_2: - Edges and flows: $u \rightarrow x_1$ (3,3), $u \rightarrow x_4$ (1,5), $x_1 \rightarrow x_2$ (3,7), $x_1 \rightarrow x_3$ (0,1), $x_2 \rightarrow v$ (3,6), $x_3 \rightarrow v$ (1,2), $x_4 \rightarrow x_3$ (1,4).

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0102	146	—	—	—	The top diagram shows a flow network with source u and sink v. Intermediate nodes are x_1, x_2, x_3, x_4. The flow F_3 is indicated. The bottom diagram shows the same network with the maximum flow F_4.
voloshin-hypergraph_ DIA_0103	151	—	—	—	A hypergraph diagram with 6 nodes and 4 hyperedges. The nodes are arranged in a hexagonal pattern, and the hyperedges are represented by closed curves connecting different subsets of nodes.
voloshin-hypergraph_ DIA_0104	155	—	—	—	Two hypergraph diagrams labeled $\mathcal{H}_1$ and $\mathcal{H}_2$. $\mathcal{H}_1$ is a smaller hypergraph with 6 nodes and 4 hyperedges. $\mathcal{H}_2$ is a larger hypergraph with 8 nodes and 6 hyperedges.

Identifier	Page	Conf.	LaTeX source	Rendered	Image																																										
voloshin-hypergraph_ DIA_0105	157	—	—	—	The diagram illustrates the relationship between a hypergraph $\mathcal{H} = (X, \mathcal{D})$, its incidence matrix $I(\mathcal{H})$, and its bipartite graph $B(\mathcal{H})$. Hypergraph $\mathcal{H}$: A set of 6 vertices $\{1, 2, 3, 4, 5, 6\}$ and 5 hyperedges D_1, D_2, D_3, D_4, D_5. Vertex 1 is in D_1, vertex 2 is in D_2, vertex 3 is in D_4, vertex 4 is in D_5, and vertex 5 is in D_3. Hyperedges D_1, D_2, D_3 are nested, while D_4 and D_5 are larger and contain the others. Incidence Matrix $I(\mathcal{H})$: <table border="1"> <thead> <tr> <th></th><th>D_1</th><th>D_2</th><th>D_3</th><th>D_4</th><th>D_5</th></tr> </thead> <tbody> <tr> <th>1</th><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr> <tr> <th>2</th><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td></tr> <tr> <th>3</th><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td></tr> <tr> <th>4</th><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td></tr> <tr> <th>5</th><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td></tr> <tr> <th>6</th><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr> </tbody> </table> Bipartite Graph $B(\mathcal{H})$: A bipartite graph with vertices $\{1, 2, 3, 4, 5, 6\}$ on the left and $\{D_1, D_2, D_3, D_4, D_5\}$ on the right. Edges connect vertices to the hyperedges they belong to: $(1, D_1), (2, D_2), (3, D_4), (4, D_5), (5, D_3)$. Hypergraph $\mathcal{H}^*$: A set of 6 vertices $\{1, 2, 3, 4, 5, 6\}$ and 5 hyperedges d_1, d_2, d_3, d_4, d_5. Vertex 1 is in d_1, vertex 2 is in d_2, vertex 3 is in d_3, vertex 4 is in d_4, and vertex 5 is in d_5. Hyperedges d_1, d_2, d_3 are nested, while d_4 and d_5 are larger and contain the others.		D_1	D_2	D_3	D_4	D_5	1	1	1	1	0	0	2	0	1	1	1	0	3	0	0	0	1	1	4	0	0	1	0	1	5	0	0	0	1	1	6	0	0	0	0	0
	D_1	D_2	D_3	D_4	D_5																																										
1	1	1	1	0	0																																										
2	0	1	1	1	0																																										
3	0	0	0	1	1																																										
4	0	0	1	0	1																																										
5	0	0	0	1	1																																										
6	0	0	0	0	0																																										
voloshin-hypergraph_ DIA_0106	158	—	—	—	The diagram illustrates the relationship between a graph G and its dual graph G^*. Graph G: A graph with 5 vertices $\{1, 2, 3, 4, 5\}$ and 5 edges a, b, c, d, e. Vertex 1 is connected to 2 by edge b, vertex 2 is connected to 3 by edge c, vertex 3 is connected to 4 by edge d, vertex 4 is connected to 5 by edge e, and vertex 5 is connected to 1 by edge a. Dual Graph G^*: A graph with 5 vertices $\{a, b, c, d, e\}$ and 5 edges $1, 2, 3, 4, 5$. Vertex a is connected to b by edge 1, vertex b is connected to c by edge 2, vertex c is connected to d by edge 3, vertex d is connected to e by edge 4, and vertex e is connected to a by edge 5.																																										

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0107	161	—	—	—	Diagrams illustrating various hypergraph types K_4^i for $i = 0, 1, 2, 3, 4$. K_4^0 is a single vertex. K_4^1 is a single edge (two vertices). K_4^2 is a square with both diagonals. K_4^3 is a complete graph K_4 with four vertices and six edges. K_4^4 is a single hyperedge containing all four vertices.
voloshin-hypergraph_ DIA_0108	163	—	—	—	Diagram illustrating the strong deletion of a vertex x from a hypergraph $\mathcal{H}$. The left side shows $\mathcal{H}$ with vertex x (circled) and its incident edges. An arrow labeled "delete x strongly" points to the resulting hypergraph $\mathcal{H}_1 = \mathcal{H} - x$, where x and its incident edges have been removed.
voloshin-hypergraph_ DIA_0109	163	—	—	—	Diagram illustrating the weak deletion of a vertex x from a hypergraph $\mathcal{H}$. The left side shows $\mathcal{H}$ with vertex x (circled) and its incident edges. An arrow labeled "delete x weakly" points to the resulting hypergraph $\mathcal{H}_1 = \mathcal{H} - x$, where x is removed but its incident edges remain as separate hyperedges.
voloshin-hypergraph_ DIA_0110	164	—	—	—	Diagram illustrating the strong deletion of a set D from a hypergraph $\mathcal{H}$. The left side shows $\mathcal{H}$ with a set D (circled) and its incident edges. An arrow labeled "delete D strongly" points to the resulting hypergraph $\mathcal{H}_1 = \mathcal{H} - D$, where D and its incident edges have been removed.

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voloshin-hypergraph_ DIA_0111	164	—	—	—	
voloshin-hypergraph_ DIA_0112	165	—	—	—	
voloshin-hypergraph_ DIA_0113	166	—	—	—	

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voloshin-hypergraph_ DIA_0114	168	—	—	—	
voloshin-hypergraph_ DIA_0115	171	—	—	—	
voloshin-hypergraph_ DIA_0116	172	—	—	—	

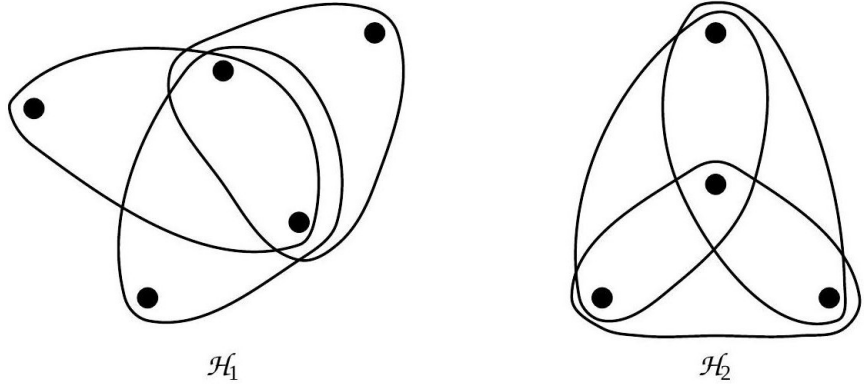
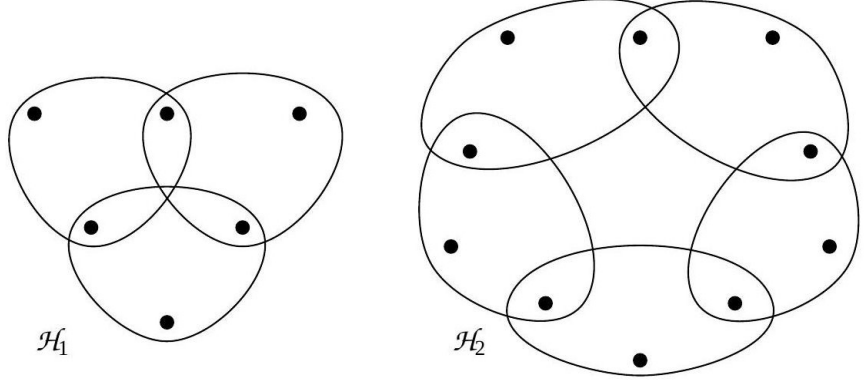
Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0117	174	—	—	—	$\mathcal{H} = K_3$ not conformal, not Helly $\mathcal{H}_1$ - conformal, not Helly $\mathcal{H}_2 \cong \mathcal{H}_1^*$ - Helly, not conformal $\mathcal{H}_3 \cong \mathcal{H}_3^*$ - Helly, conformal
voloshin-hypergraph_ DIA_0118	178	—	—	—	T C_4 T C_4

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0119	179	—	—	—	$\mathcal{H}_1$ $\mathcal{H}_2$
voloshin-hypergraph_ DIA_0120	180	—	—	—	$\mathcal{H}$ $\mathcal{H}_0$
voloshin-hypergraph_ DIA_0121	181	—	—	—	$\mathcal{H}_0^*$ $\mathcal{H}^*$
voloshin-hypergraph_ DIA_0122	181	—	—	—	$\mathcal{H}_0^*$ $\mathcal{H}^*$
voloshin-hypergraph_ DIA_0123	182	—	—	—	G $\mathcal{H}$ $\mathcal{H}^*$

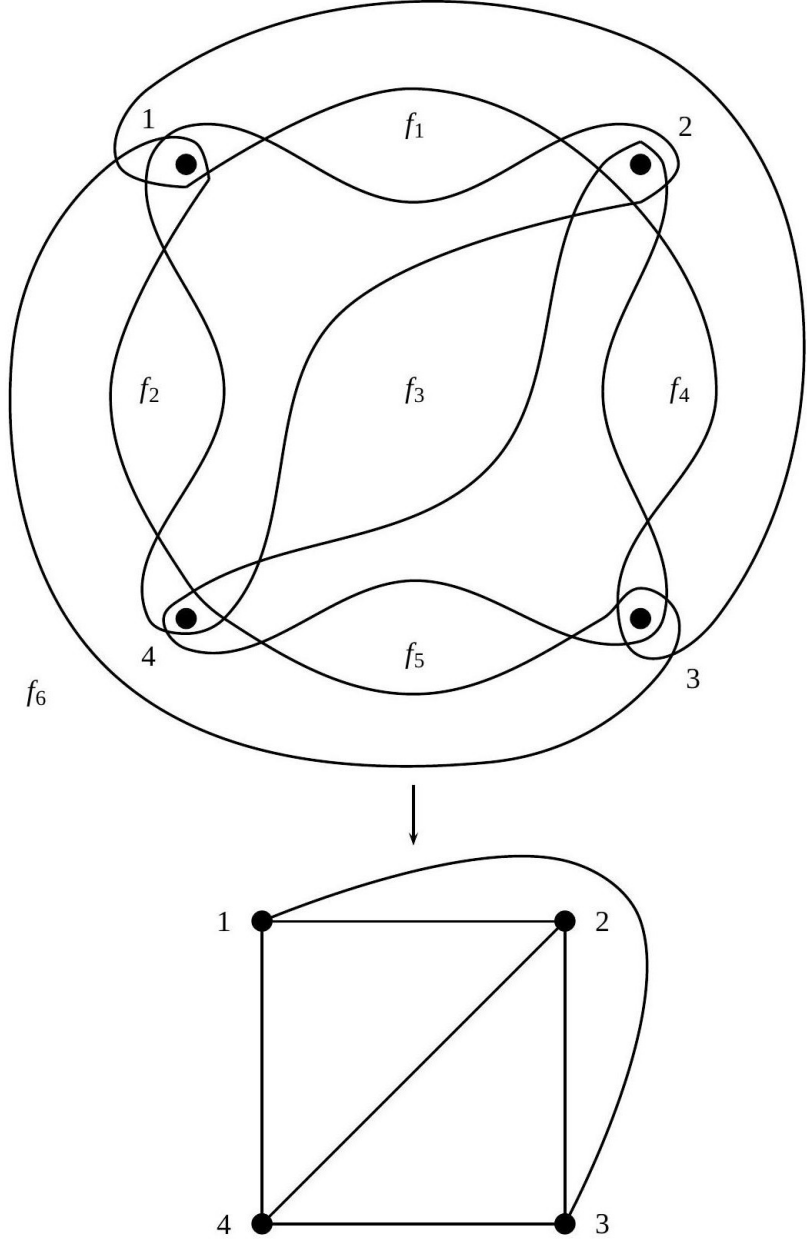
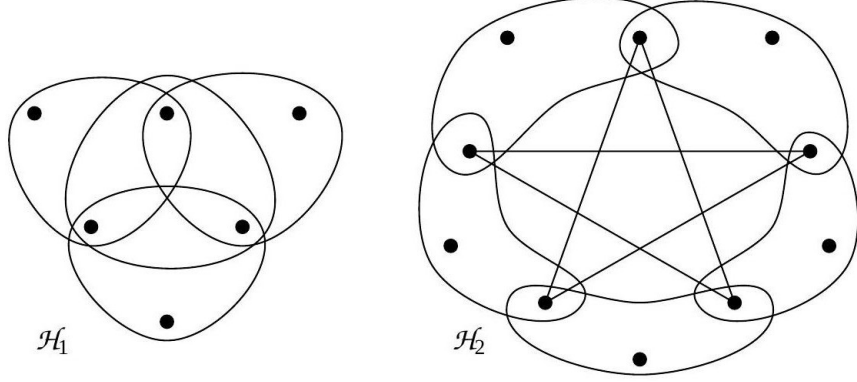
Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0124	183	—	—	—	
voloshin-hypergraph_ DIA_0125	186	—	—	—	
voloshin-hypergraph_ DIA_0126	187	—	—	—	

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voloshin-hypergraph_ DIA_0127	188	—	—	—	
voloshin-hypergraph_ DIA_0128	189	—	—	—	
voloshin-hypergraph_ DIA_0129	191	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0130	194	—	—	—	Diagram illustrating a hypergraph $\mathcal{H}$ and its relationship to a tree $T \subseteq [\mathcal{H}]_2$. The top part shows $\mathcal{H}$ with a vertex x and an arrow pointing to $T \subseteq [\mathcal{H}]_2$. The bottom part shows $\mathcal{H} - x$ and an arrow pointing to $T - x$.
voloshin-hypergraph_ DIA_0131	196	—	—	—	Diagram illustrating two hypergraphs $\mathcal{H}_1$ and $\mathcal{H}_2$.
voloshin-hypergraph_ DIA_0132	198	—	—	—	Diagram illustrating a hypergraph structure with multiple vertices and hyperedges.
voloshin-hypergraph_ DIA_0133	199	—	—	—	Diagram illustrating two hypergraph structures, one above the other.

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voloshin-hypergraph_ DIA_0134	200	—	—	—	
voloshin-hypergraph_ DIA_0135	202	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0136	203	—	—	—	

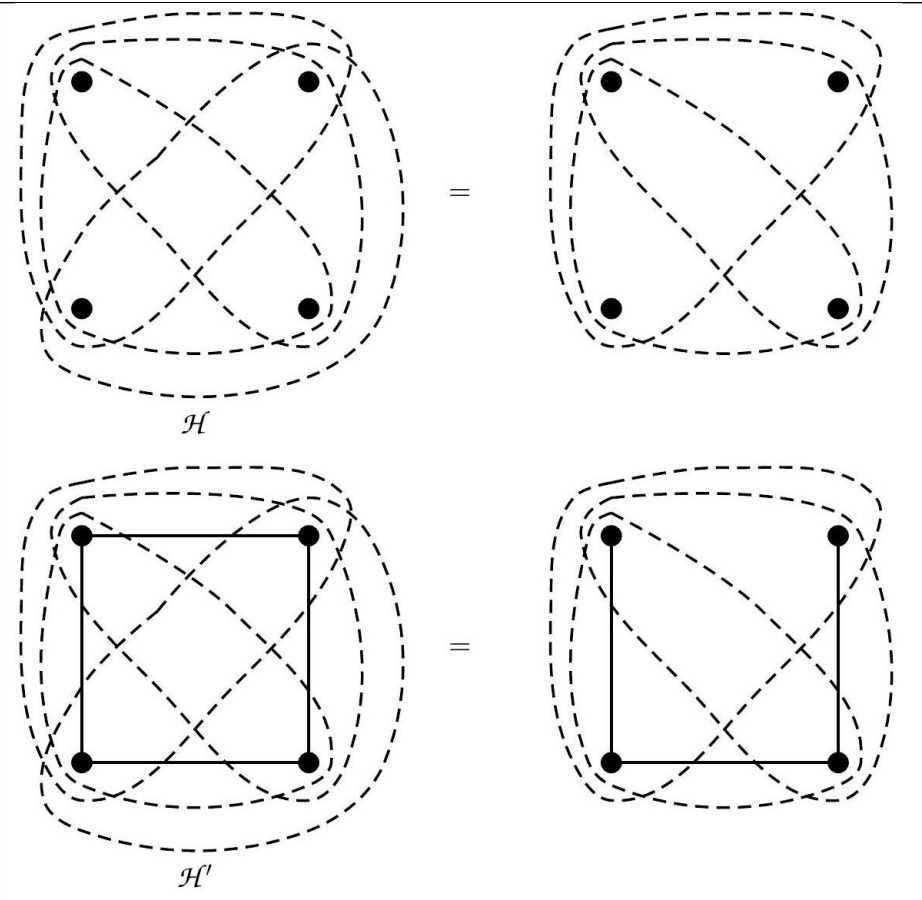
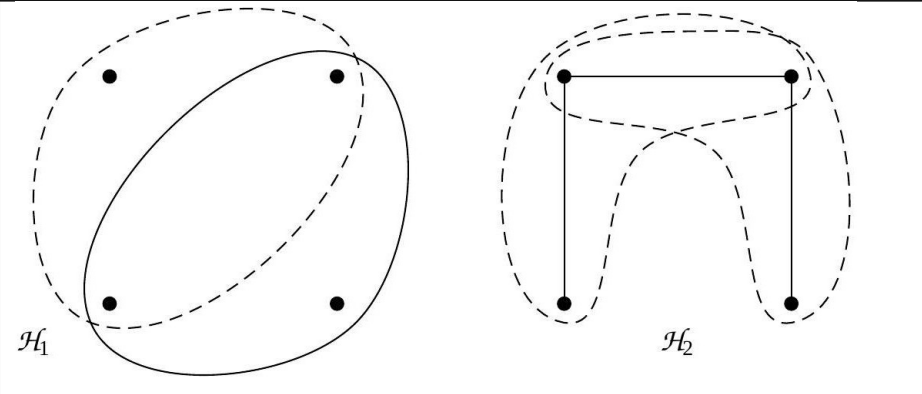
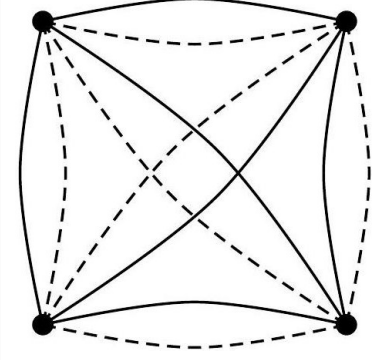
Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0137	206	—	—	—	 The diagram illustrates a hypergraph with four vertices labeled 1, 2, 3, and 4. Six hyperedges, labeled f_1 through f_6, are shown as closed loops connecting these vertices. An arrow points down to a graph representation where the vertices are arranged in a square (1 at top-left, 2 at top-right, 3 at bottom-right, 4 at bottom-left). The edges of this graph correspond to the hyperedges: f_1 is the top edge (1-2), f_2 is the left edge (1-4), f_3 is the diagonal (1-3), f_4 is the right edge (2-3), f_5 is the bottom edge (3-4), and f_6 is a curved edge connecting 1 and 2.
voloshin-hypergraph_ DIA_0138	207	—	—	—	 Two hypergraphs are shown side-by-side. The left one, labeled $\mathcal{H}_1$, has six vertices and three hyperedges represented by overlapping loops. The right one, labeled $\mathcal{H}_2$, has six vertices arranged in a circle, with hyperedges forming a complex internal structure including a star-like pattern.

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voloshin-hypergraph_ DIA_0139	210	—	—	—	
voloshin-hypergraph_ DIA_0140	213	—	—	—	
voloshin-hypergraph_ DIA_0141	216	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0142	217	—	—	—	
voloshin-hypergraph_ DIA_0143	218	—	—	—	
voloshin-hypergraph_ DIA_0144	221	—	—	—	
voloshin-hypergraph_ DIA_0145	222	—	—	—	
voloshin-hypergraph_ DIA_0146	223	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0147	226	—	—	—	
voloshin-hypergraph_ DIA_0148	229	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0149	232	—	—	—	The diagram illustrates the decomposition of a hypergraph C into a sum of simpler hypergraphs through a series of operations: Top Row: Hypergraph C (a diamond shape with nodes 1, 2, 3, 4 and edges D_1, D_2, D_3, D_4) is shown. It is equal to the sum of two hypergraphs: one with a vertical edge D_1 and one with a vertical edge D_2. This step is labeled "splitting 13". Second Row: The two hypergraphs from the top row are shown. The first is labeled "elimination" and the second is labeled "C-clearing + contraction 24". They are equal to the sum of two hypergraphs: one with a vertical edge D_1 and one with a vertical edge D_2. This step is labeled "spl. 24". Third Row: The two hypergraphs from the second row are shown. The first is labeled "elimination" and the second is labeled "C-clearing + contraction 24". They are equal to the sum of two hypergraphs: one with a vertical edge D_1 and one with a vertical edge D_2. This step is labeled "spl. 24". Bottom Row: The two hypergraphs from the third row are shown. The first is labeled "elimination" and the second is labeled "C-clearing + contraction 24". They are equal to the sum of two hypergraphs: one with a vertical edge D_1 and one with a vertical edge D_2. This step is labeled "contraction 24".

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voloshin-hypergraph_ DIA_0150	234	—	—	—	
voloshin-hypergraph_ DIA_0151	235	—	—	—	
voloshin-hypergraph_ DIA_0152	237	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0153	238	—	—	—	
voloshin-hypergraph_ DIA_0154	239	—	—	—	
voloshin-hypergraph_ DIA_0155	242	—	—	—	
voloshin-hypergraph_ DIA_0156	242	—	—	—	$\mathcal{H}$

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voloshin-hypergraph_ DIA_0157	244	—	—	—	
voloshin-hypergraph_ DIA_0158	250	—	—	—	
voloshin-hypergraph_ DIA_0159	251	—	—	—	
voloshin-hypergraph_ DIA_0160	253	—	—	—	

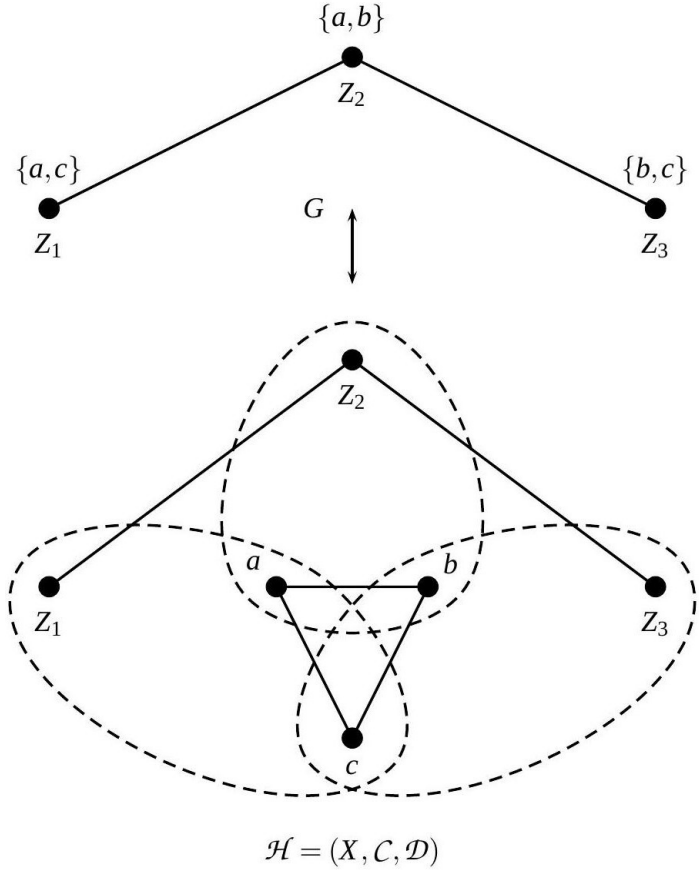
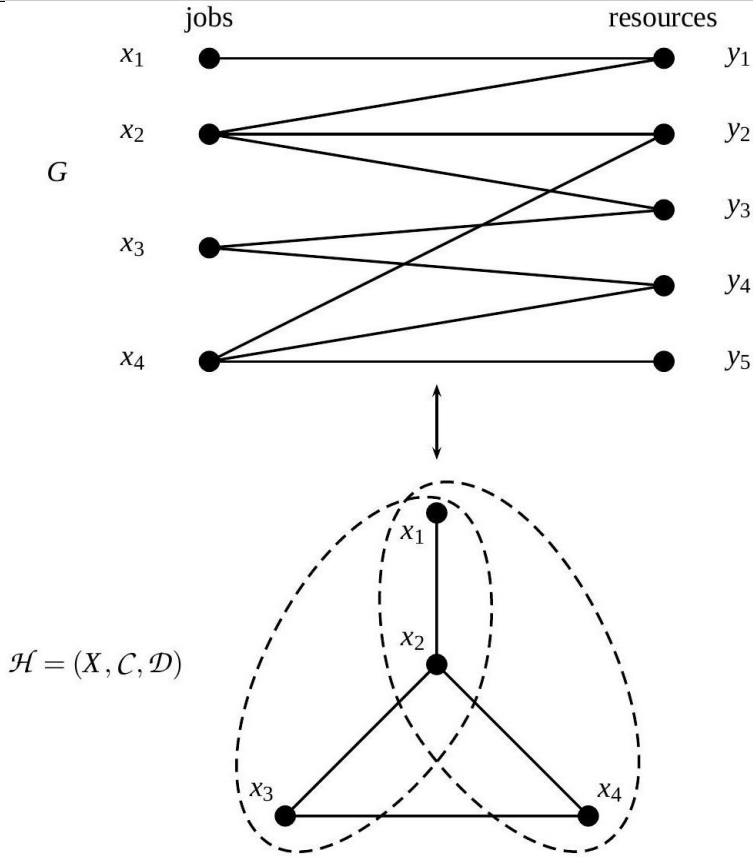
Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0161	257	—	—	—	
voloshin-hypergraph_ DIA_0162	258	—	—	—	
voloshin-hypergraph_ DIA_0163	259	—	—	—	

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voloshin-hypergraph_ DIA_0164	261	—	—	—	
voloshin-hypergraph_ DIA_0165	262	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0166	265	—	—	—	
voloshin-hypergraph_ DIA_0167	267	—	—	—	
voloshin-hypergraph_ DIA_0168	269	—	—	—	

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voloshin-hypergraph_ DIA_0169	271	—	—	—	
voloshin-hypergraph_ DIA_0170	275	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0171	276	—	—	—	
voloshin-hypergraph_ DIA_0172	277	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0173	280	—	—	—	 $\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$
voloshin-hypergraph_ DIA_0174	281	—	—	—	 $\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0175	284	—	—	—	Induction basis Induction hypothesis Induction step 1. P_1 2. $P_k \rightarrow P_{k+1}$ $\longrightarrow$ P_n is true for all $n = 1, 2, \dots$
voloshin-hypergraph_ DIA_0176	284	—	—	—	Induction basis Induction hypothesis Induction step 1. P_1 2. $P_1, P_2, \dots, P_k \rightarrow P_{k+1}$ $\longrightarrow$ P_n is true for all $n = 1, 2, \dots$